

3. Data Inventory and Features Extracted

❖ Section 1: Data Dictionary

DS01: World Bank Commodity Price Data (Pink Sheet)

URL <https://www.worldbank.org/en/research/commodity-markets>

Proxy Relationship: Tracks chemical and energy commodity prices that determine upstream API manufacturing costs. Direct API price data is not publicly available, so this is the closest rigorous proxy.

Key data points extracted: Crude oil (Brent) up 49% from 2024 to April 2026. Urea up 153% over the same period. Natural gas (Europe) up 41%. Aluminium up 49%. These commodity spikes directly translate to API manufacturing cost pressure and packaging cost increases, the upstream conditions that precede drug shortages.

DS02: IDSP Weekly Outbreak Surveillance Data

Key insight for model: IDSP operates a five-tier trigger system. A trigger level of 3 or above confirms an outbreak at the district or state level and activates the demand spike signal in our model. An average of 40–50 outbreaks reported nationally per week provides a continuous stream of signals.

DS03: GeM Procurement Data and Research

Key data point: Of 1507 medicines demanded through GeM by a government hospital, only 695 (46.1%) were successfully procured (a 53.9% shortfall). Procurement cycle spans 5–6 months end to end.

DS04: FDA Drug Shortage Database

URL <https://www.accessdata.fda.gov/scripts/drugshortages/>

Geographic Limitation: US market only. Not used as a direct India signal source.

Key validation finding: Every drug category flagged as high-risk in the Pharma-Pulse model (antibiotic injections, ICU medicines, oncology drugs, IV fluids) appears in the FDA active shortage list independently. Cross-market convergence validates model drug prioritisation.

DS05: CDSCO Drug Approval and Recall Database

Key insight: WHO (2016) confirmed that long regulatory approval timelines make market entry costly and unpredictable, directly contributing to shortage risk. CDSCO backlog data quantifies this signal for India.

DS06: News Source-FOPE Warning, New Indian Express and TMH Nurse Interview.

Real-world validation: Confirms geopolitical disruption signal layer. Drugs named in the article match four of our six highest-risk dashboard drugs.

Key finding: FOPE formally warned the Department of Pharmaceuticals of imminent shortage of Paracetamol, Amoxicillin, Metformin, and Azithromycin due to West Asia conflict-driven API scarcity. These are exactly the drugs Pharma-Pulse flags as high-risk. The upstream signals were visible 8–10 weeks before this warning was issued.

❖ Section 2: Feature Definition

Model type: Hybrid weighted composite + Random Forest probability layer.

Output: Shortage risk score per drug per region, scale 0–100.

Scoring formula: Final Score = $(0.6 \times \text{Composite Score}) + (0.4 \times \text{RF Probability} \times 100)$

Feature F-01: API Price Change (4-Week Rolling)

Calculation:

Step 1: Extract monthly prices for four proxy commodities.

Step 2: Calculate weighted average composite index (crude oil 35%, urea 30%, natural gas 20%, aluminium 15%).

Step 3: Calculate 4-week rolling percentage change from the previous period.

Step 4: Normalise to a 0–1 scale.

Threshold for alert: Greater than 15% increase over a 4-week window triggers an elevated signal.

Feature F-02: Supplier HHI Concentration Score

Calculation:

Step 1: For each drug category, extract the count of unique vendors who submitted valid bids in the last 3 tender cycles.

Step 2: Calculate each vendor's market share as the proportion of total bids.

Step 3: HHI = sum of squared market shares.

Step 4: Scale HHI (0 to 1), where 1 = single-supplier monopoly.

Threshold for alert: HHI greater than 0.60 triggers an elevated signal for fewer than 3 effective suppliers.

Feature F-03: CDSCO Regulatory Backlog

Calculation:

Step 1: Download CDSCO approval list and recall notices monthly.

Step 2: Count pending approvals per drug category.

Step 3: Count active recalls per category.

Step 4: Composite backlog score = (pending approvals × 0.6) + (active recalls × 0.4).

Step 5: Normalise to a 0–1 scale.

Threshold for alert: Any active recall in a drug category automatically elevates the risk score by 0.2 regardless of other features.

Feature F-04: Disease Outbreak Signal

Calculation:

Step 1: Download weekly IDSP bulletin.

Step 2: Extract outbreak type and IDSP trigger level per state.

Step 3: Map outbreak type to drug demand category using outbreak-drug mapping table (Section 3).

Step 4: Weight signal by trigger level: Trigger 1 = 0.2, Trigger 2 = 0.5, Trigger 3 = 0.8, Trigger 4 = 1.0.

Step 5: Aggregate by state and drug category.

Step 6: Normalise to a 0–1 scale.

Threshold for alert: Trigger level 3 or above in any state for a mapped drug category activates demand spike flag.

Feature F-05: China API Export Volume Delta

Calculation:

Step 1: Extract relevant World Bank trade indices monthly.

Step 2: Calculate month-on-month percentage change.

Step 3: Supplement with binary geopolitical disruption flag: 0 = no active restriction, 1 = confirmed restriction event.

Step 4: Combined score = (trade index delta × 0.7) + (geopolitical flag × 0.3).

Step 5: Normalise to a 0–1 scale.

Threshold for alert: Greater than 20% export volume decline or any confirmed restriction event triggers an elevated signal.

Feature F-06: GeM Tender Fill Rate

Calculation:

Step 1: Extract GeM procurement reports by drug category.

Step 2: Calculate fill rate = orders materialised / orders placed.

Step 3: Calculate 3-quarter rolling average fill rate.

Step 4: Invert for risk score (low fill rate = high risk).

Step 5: Normalise to a 0–1 scale.

Threshold for alert: Rolling fill rate below 60% for a drug category triggers an elevated signal.

Feature Summary Table

ID	Feature Name	Weight	Source	Direction	Alert Threshold
F-01	API price change 4-week rolling	0.28	DS-01 World Bank	↑ price = ↑ risk	>15% increase
F-02	Supplier HHI concentration	0.22	DS-03 GeM	↑ HHI = ↑ risk	HHI >0.60
F-03	CDSCO regulatory backlog	0.18	DS-05 CDSCO	↑ backlog = ↑ risk	Any active recall
F-04	Disease outbreak signal	0.16	DS-02 IDSP	↑ outbreaks = ↑ risk	Trigger level ≥3
F-05	China export volume delta	0.10	DS-01 + DS-09	↓ exports = ↑ risk	>20% decline
F-06	GeM tender fill rate	0.06	DS-03 GeM	↓ fill rate = ↑ risk	Fill rate <60%
—	Total weight	1.00	—	—	—

Weight Justification:

The weights were derived through three inputs working together.

First, WHO (2016) and the IJMR paper both identify supply-side factors (API availability and supplier concentration) as the primary structural drivers of shortage, which is why F-01 and F-02 carry the two highest weights.

Second, regulatory delays and demand spikes are confirmed secondary drivers by multiple sources, giving F-03 and F-04 their mid-range weights.

Third, China export volume and GeM fill rate are real but more volatile and harder to measure precisely, so they carry lower weights to avoid model instability from noisy signals.

The 0.6 composite and 0.4 Random Forest split means that my model is always explainable. The composite layer can be broken down into plain English for any procurement officer. The RF layer adds predictive power without sacrificing interpretability.

❖ Section 3: Outbreak-to-Drug Demand Mapping

This table defines the translation layer between IDSP outbreak type and predicted demand surge for specific drug categories. When an IDSP bulletin reports an outbreak, the model uses this table to identify which drugs face elevated demand pressure in the affected state and adjusts their risk scores accordingly.

Disease	Trigger	Surge Multiplier	Alert Window	Key Drugs at Risk
Cholera	T2	2.4×	2 Weeks	ORS, IV Fluids, Doxycycline
Dengue	T2	2.1×	2 Weeks	Paracetamol, IV Fluids, Platelet Support
ARI / Influenza	T2	2.6×	1 Week	Amoxicillin, Azithromycin, Oseltamivir
JE / AES	T2	2.8×	1 Week	Mannitol, Phenytoin, Diazepam
COVID-19 / SARI	T3	3.2×	1 Week	Remdesivir, ICU Sedatives, Azithromycin
Unusual Syndrome	T3	2.0×	Immediate	Broad-Spectrum Antibiotics, Critical Care Drugs

(The table is for representation purposes; actual data is much bigger.)

"When an IDSP bulletin reports a dengue outbreak at Trigger Level 2 in Maharashtra, Pharma-Pulse automatically elevates the risk score for Paracetamol IV, IV fluids, and Ondansetron in Maharashtra by the corresponding demand multiplier two weeks before that demand surge reaches a hospital pharmacy shelf."

```

# For each outbreak entry in the IDSP bulletin:

outbreak_type = "Acute respiratory illness"
state = "Maharashtra"
trigger_level = 3

# Look up mapping table
mapped_drugs = get_drugs_for_outbreak(outbreak_type)
# Returns: ['Amoxicillin', 'Azithromycin', 'Oseltamivir',
#          'Paracetamol', 'Salbutamol', 'Prednisolone']

# Get demand multiplier
multiplier = get_multiplier(outbreak_type)
# Returns: 2.6

# Get signal weight by trigger level
trigger_weight = {1: 0.2, 2: 0.5, 3: 0.8, 4: 1.0}[trigger_level]
# Returns: 0.8

# Calculate outbreak signal contribution
outbreak_signal = min(multiplier * trigger_weight / 3.2, 1.0)
# Normalised to 0-1

# Apply to risk score for each mapped drug in Maharashtra
for drug in mapped_drugs:
    risk_scores[drug]['Maharashtra']['outbreak_signal'] = outbreak_signal

```

(When the model ingests a weekly IDSP bulletin, it runs this logic.)

When multiple outbreaks are active simultaneously in the same state, which happens during monsoon season, the model does not simply add signals. It uses a dampened aggregation to avoid score inflation:

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python

# If multiple outbreaks affect the same drug in the same state:
# Final signal = highest single outbreak signal
#               + 0.3 × sum of remaining signals
# Capped at 1.0

# Example: Maharashtra has both Dengue and ARI active
# Dengue signal for Paracetamol = 0.84
# ARI signal for Paracetamol = 0.92 (higher)
# Final = 0.92 + (0.3 × 0.84) = 0.92 + 0.25 = 1.0 (capped)

```

Seasonal Pattern Layer

Based on IDSP historical data patterns going back to 2008, certain outbreak types follow predictable seasonal cycles. This allows the model to apply a forward-looking seasonal multiplier even before an outbreak is officially reported.

Season	Peak Period	High-Risk Outbreak Types	Pre-Alert Multiplier
Pre-monsoon	March — May	Heat stroke, ARI, Food poisoning	1.2×
Monsoon	June — September	Dengue, Malaria, AWD, Cholera, Hepatitis A/E	1.8×
Post-monsoon	October — November	Dengue (peak), JE, Scrub typhus	1.6×
Winter	December — February	ARI, Influenza, Pneumonia	1.4×

The seasonal multiplier activates automatically based on calendar date, even without an active IDSP outbreak report. This means the model begins elevating risk scores for monsoon-linked drugs in late May, two to three weeks before the first outbreak reports typically arrive.

This is the only forward-looking element in your model that does not require a data input to trigger. It runs on the calendar. This is what you explain when someone asks how your model provides early warning before the data even arrives.

Drug Convergence Analysis (Most Vulnerable Drugs)

Reading across the entire mapping table, some drugs appear in multiple outbreak categories. These are your most structurally vulnerable drugs because any one of several outbreak types can spike their demand simultaneously.

Drug	Number of Outbreak Types That Spike Demand	Risk Implication
Paracetamol IV	8 out of 14 outbreak types	Highest multi-signal vulnerability — always on watch list
IV fluids (NS / RL)	10 out of 14 outbreak types	Universal shortage risk across all outbreak seasons
Amoxicillin injection	5 out of 14 outbreak types	Chronic shortage risk confirmed by FDA, FOPE, TMH interview
Azithromycin	4 out of 14 outbreak types	Second line for multiple infections — substitution pressure
Ondansetron injection	3 out of 14 outbreak types	ICU and acute care — low margin for substitution

This convergence analysis is what builds your dashboard drug priority list. It is not arbitrary. Every drug on your dashboard is there because this table says it should be.

❖ Section 5: Data Architecture Diagram

The architecture operates across four layers.

Layer 1 ingests six data sources: World Bank commodity prices, IDSP weekly outbreak bulletins, GeM procurement data, CDSCO regulatory records, FDA shortage data for validation, and primary research from TMH Jamshedpur.

Layer 2 is the Python processing pipeline where data is cleaned, features are engineered, and the outbreak-to-drug mapping from Section 3 is applied. Six features emerge from this layer, each with a documented weight and calculation method as defined in Section 2. The seasonal pattern layer operates independently of incoming data.

Layer 3 combines the six features into a weighted composite score, then blends it with a Random Forest probability layer in a 60/40 ratio to produce the final shortage risk score per drug per region.

Layer 4 is the output: the dashboard delivering the risk heatmap, drug drilldown, procurement recommendations, and scenario simulator to the end user.

The critical design principle: Every layer is transparent and traceable. Any risk score can be decomposed back to its feature contributions, which can be traced back to their source data, which can be traced back to their original public dataset. No black box. Full audit trail from raw data to procurement recommendation.

